DEEP LEARNING BASED DETECTION OF SMS PHISING ATTACK

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INTRODUCTION

Smishing, or SMS phishing, is a cyberattack where fraudulent text messages trick users into revealing sensitive personal and financial information. With rising smartphone usage worldwide, smishing has become an increasingly sophisticated threat, exposing millions of users to financial loss, identity theft, and privacy breaches.

MOTIVATION

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- Smishing messages are constantly evolving, making detection a moving target.
- SMS language is informal, filled with slang and abbreviations, adding complexity to detection.
- Balancing detection accuracy while minimizing false alarms remains an ongoing challenge.

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PROBLEM STATEMENT

Current smishing detection systems mainly focus on classifying messages as phishing or legitimate, without identifying the specific attack type. In addition, many models are either too complex for real-time mobile use or can be affected by simple evasion techniques. This creates a need for a lightweight and efficient system that can detect phishing messages and also identify the attack type in a practical setting.

LITERATURE REVIEW

TITLE	METHODOLOGY	INFERENCES
[1] Mehmood, Muhammad Khalid, Humaira Arshad, Moatsum Alawida, and Abid Mehmood. "Enhancing smishing detection: A deep learning approach for improved accuracy and reduced false positives." IEEE Access 12 (2024): 137176-137193.	Proposes a CNN-LSTM hybrid model trained on two merged English SMS datasets. Preprocessing includes tokenization, stopword removal, and lemmatization.Benchmarked against ML and DL baselines.	1. Achieves 99.74% accuracy. 2. Significantly reduces false positives compared to other models.
[2] Salman, Muhammad, Muhammad Ikram, and Mohamed Ali Kaafar. "Investigating evasive techniques in sms spam filtering: A comparative analysis of machine learning models." IEEE Access 12 (2024): 24306-24324.	Benchmarks shallow ML and deep learning models on a large English SMS dataset of 67,018 messages. Evaluates robustness against six evasion techniques.	1. RoBERTa achieved the best F1-score of 98%. 2. All models are vulnerable to evasion techniques, with spacing being the most effective attack.

LITERATURE REVIEW

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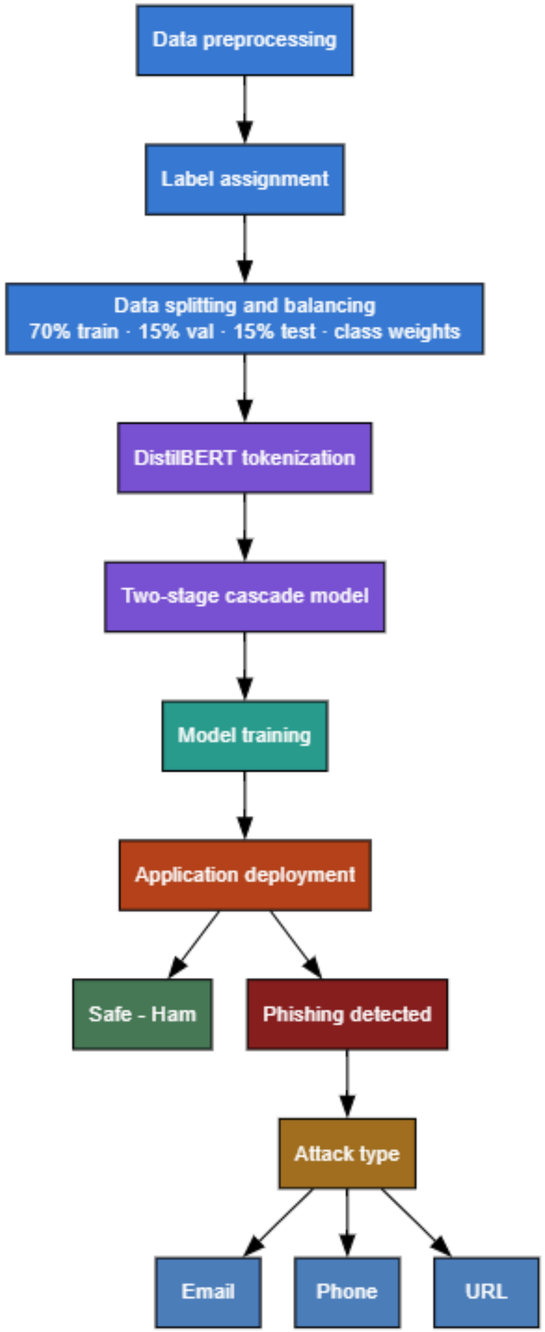
[3] Rifat, Nafiz, Mostofa Ahsan, Md Chowdhury, and Rahul Gomes. "Bert against social engineering attack: Phishing text detection." In 2022 IEEE international conference on electro information technology (eIT), pp. 1-6. IEEE, 2022.	Proposes a DistilBERT-based model fine-tuned for SMS phishing text detection. Utilizes transfer learning with a pre-trained transformer architecture for binary classification of spam and legitimate messages, evaluated using standard metrics.	1. DistilBERT achieved ~99.75% accuracy, showing strong performance on short-text classification 2. Maintains high accuracy with lower computational cost compared to BERT and RoBERTa
[4] Liu, Xiaoxu, Haoye Lu, and Amiya Nayak. "A spam transformer model for SMS spam detection." IEEE Access 9 (2021): 80253-80263.	Proposes a modified Transformer model using GloVe embeddings, trained on the SMS Spam Collection v.1 dataset. Benchmarked against LR, NB, RF, SVM, LSTM, and CNN-LSTM.	1. Achieved the best accuracy of 98.92% and F1-score of 0.9613. 2. Outperformed all baseline models including CNN-LSTM on both SMS and Twitter datasets.

RESEARCH GAP

- Existing models mainly perform smishing detection without identifying attack type.
- Models struggle against real-world obfuscation techniques like hidden symbols or spacing.
- Many models are heavy for real-time deployment on mobile devices.

METHODLOGY


Data preprocessing
Label assignment
Data splitting and balancing
70% train · 15% val · 15% test · class weights
DistilBERT tokenization
Two-stage cascade model
Model training
Application deployment
Safe - Ham
Phishing detected
Attack type
Email
Phone
URL

DATASET

- Combined dataset from SMS Smishing Collection (Mendeley) and SmishTank.
- Total of approximately 9,300 SMS messages.
- Includes categories: Ham, Smishing, and Spam
- Features include message text, URL presence, email presence, and phone.

DATA PREPROCESSING

- **Stripping**: Remove unnecessary spaces or characters from the text
- **Lowercase Conversion**: Convert all text to lowercase for uniformity
- **Binary Encoding**: Convert Yes/No values into 1/0

LABEL ASSIGNMENT

- Each message is assigned two labels : S1 and S2
- **S1**: 0 = Ham, 1 = Phishing.
- **S2**: 0 = Phone, 1 = URL, 2 = Email Phishing .
- S2 activates only when S1 predicts Phishing.

DATA SPLITTING & CLASS BALANCING

- Dataset split into 80% Train, 10% Validation, 10% Test
- Stratified split ensures equal class distribution across all splits
- Class weights are assigned to handle imbalanced data
- Weighted Random Sampler is used during training to avoid bias

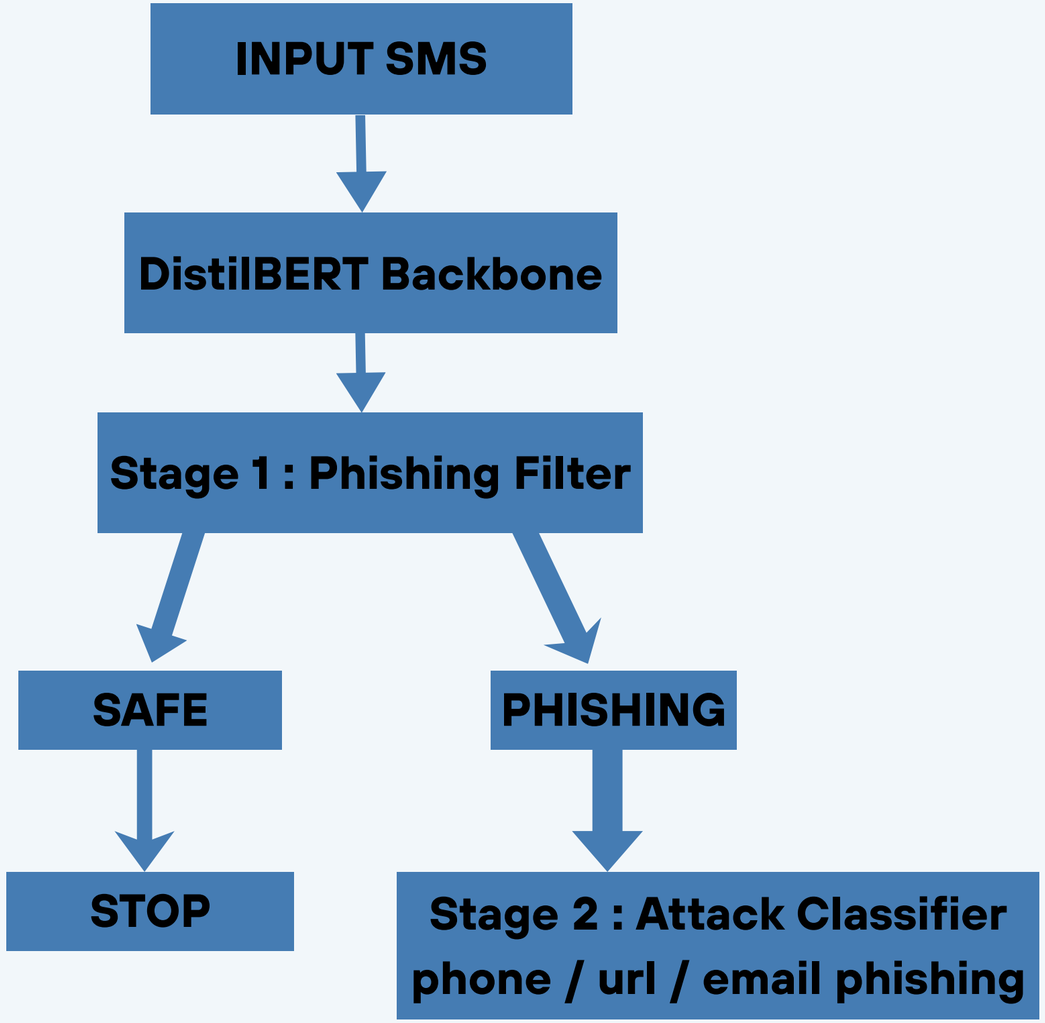
TOKENIZATION

- Each SMS is tokenized using DistilBERT's tokenizer
- [CLS] token marks the beginning, [SEP] marks the end of each message
- Max token length set to 128, shorter messages are padded with 0s
- Attention mask = 1 for real tokens, 0 for padding
- DistilBERT produces 768-dimensional vectors for each token
- The [CLS] token vector is used as the compressed representation of the entire message

PROPOSED TWO-STAGE PHISHING DETECTION FRAMEWORK

- A single DistilBERT backbone is shared across both stages
- Stage 1 detects whether a message is Ham or Phishing. The input of 768 dimensions is reduced to 128, passed through GELU activation, followed by a 30% dropout, and classified into 2 classes
- Stage 2 identifies the attack type as Phone, URL, or Email phishing. The input of 768 dimensions is reduced to 256, then to 64, and classified into 3 classes
- Stage 2 activates only when Stage 1 predicts the message as Phishing

PROPOSED FRAMEWORK



MODEL TRAINING

- The model is trained using a multi-task loss combining Stage 1 and Stage 2 losses with weights of 1.0 and 1.5 respectively
- AdamW optimizer is used with a lower learning rate for the backbone and a higher learning rate for the classification heads
- The model is trained for 6 epochs with a batch size of 32
- The best model is saved based on the combined F1 score of both stages

APPLICATION DEPLOYMENT

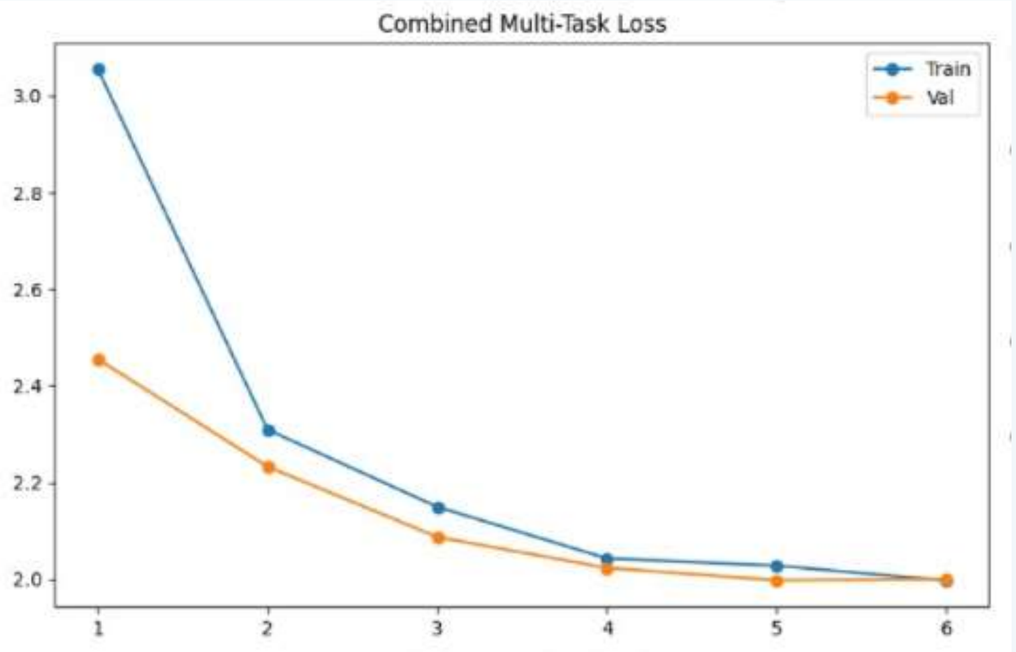
- The trained model is integrated into a mobile application built using Android Studio
- The application allows users to input an SMS message and receive an instant prediction
- The result displays whether the message is Safe or Phishing, and if phishing, identifies the attack type as Phone, URL, or Email
- The system provides a real-time, user-friendly interface for smishing detection on mobile devices

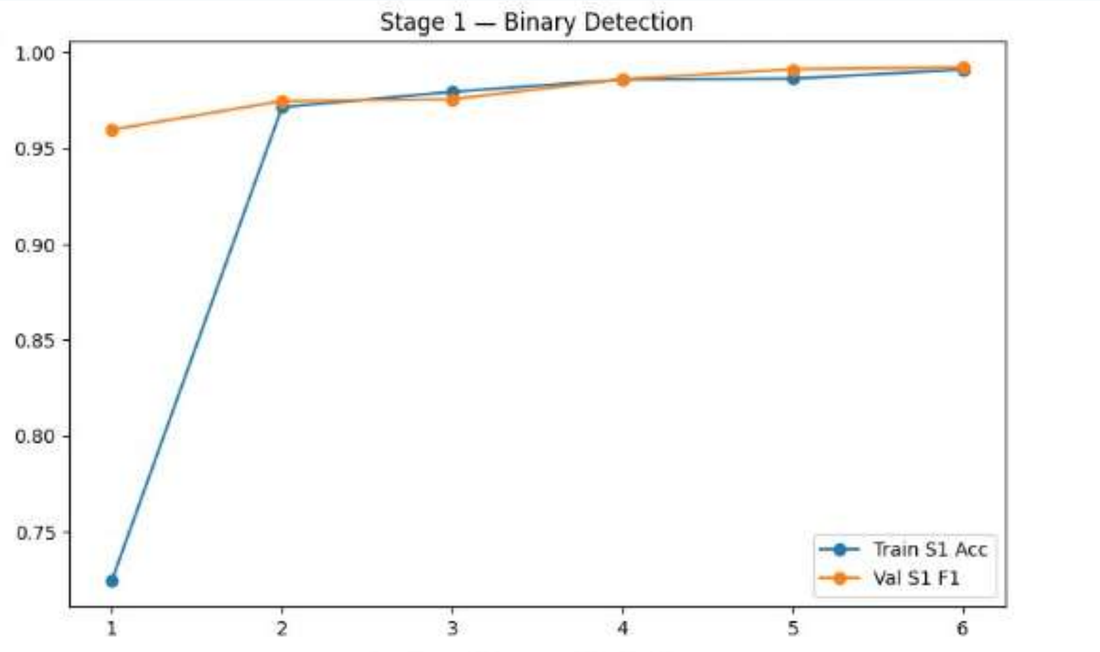
RESULTS AND DISCUSSION

- The model showed rapid convergence, with the training loss decreasing significantly across epochs.
- Stage 1 achieves high accuracy of approximately 99%, indicating strong performance in binary phishing detection.
- Stage 2 attains an accuracy of around 95%, reflecting effective classification of different attack types.
- The combined F1-score reaches approximately 0.97, demonstrating strong overall performance of the cascaded model.
- Overall, the model exhibits effective multi-task learning with consistent performance across both stages.

RESULTS AND DISCUSSION

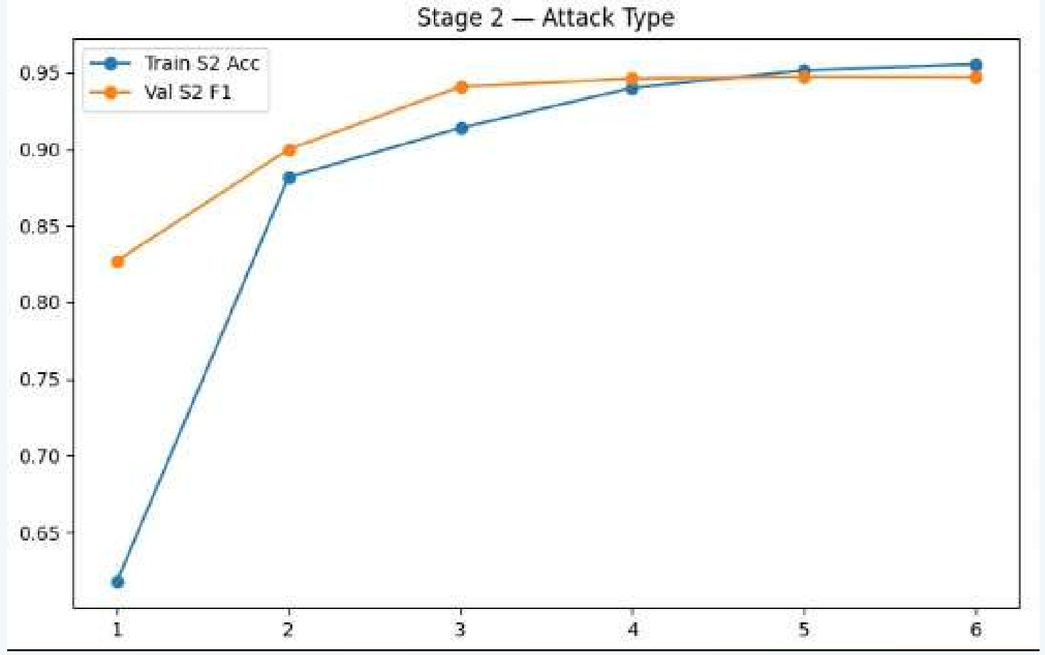
- The model shows steady convergence with decreasing loss and closely aligned training–validation curves. Stage 1 achieves high accuracy early, indicating strong phishing detection performance.

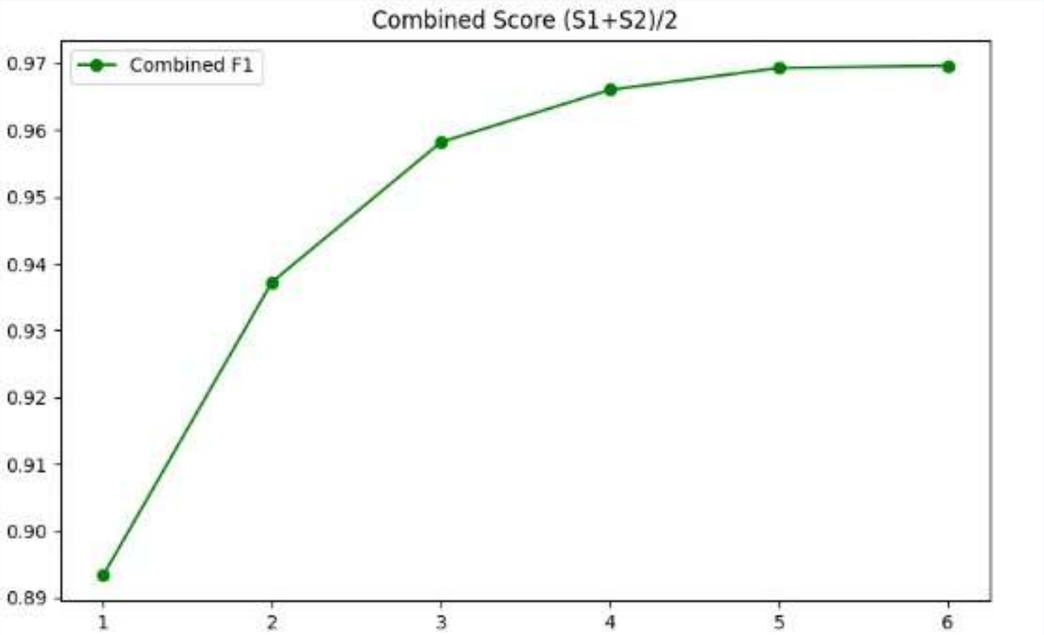




RESULTS AND DISCUSSION

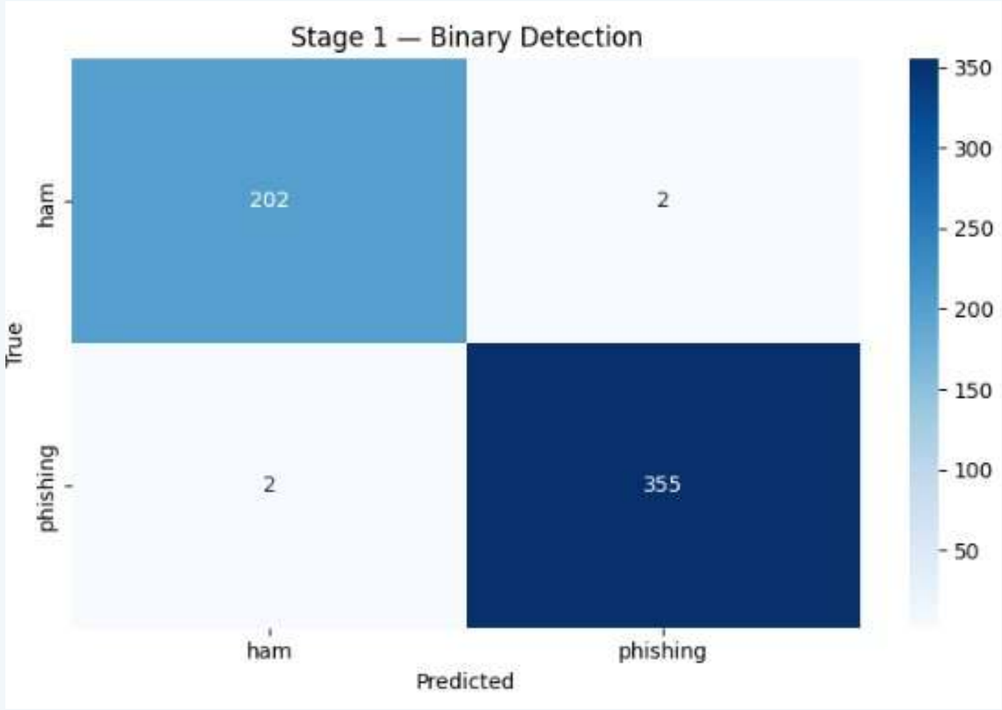
- Both stages achieve peak performance early and sustain it across epochs, demonstrating a reliable cascaded system.

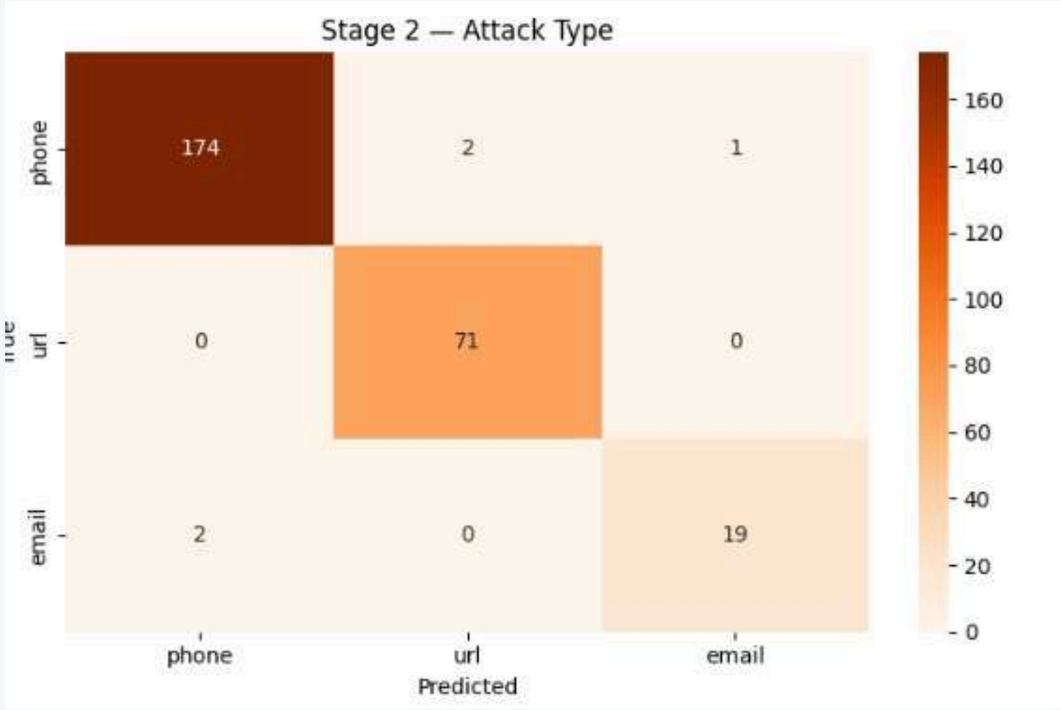




RESULTS AND DISCUSSION

- The confusion matrices show high classification accuracy in both stages, with very few misclassifications, indicating strong and reliable model performance.





FUTURE SCOPE

- Enhance the model to detect phishing in hybrid languages (like "Hinglish") and regional dialects to protect a wider range of users.
- Expand protection beyond SMS to scan for malicious links and messages on popular apps like WhatsApp and Instagram.
- Sync with a central database to automatically flag and report phishing numbers to authorities, helping to block scammers at the source.

REFERENCES

[1] Mehmood, Muhammad Khalid, Humaira Arshad, Moatsum Alawida, and Abid Mehmood. "Enhancing smishing detection: A deep learning approach for improved accuracy and reduced false positives." IEEE Access 12 (2024): 137176-137193.

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THANK YOU